

WBSU Semester II VAC Syllabus

VAC-2

Introduction to Cyber Security

Course Objectives :

1. To inform about the significance of cyber crime, tools and methods regarding security aspects.
 2. To instruct about social networking, operating system security and device security best practices.
 3. To introduce Web Application Security Methodologies and Strategic Management.
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Module I : Introduction to Cyber Crime, Tools and Security Aspects

- Overview of Computer and Web Technology
- Origin of Internet and World Wide Web
- Cyber Space and Cyber Crime
- Cyber Challenges
- Information and Data Security
- Ethical Hacking
- Ethics and Cyber Laws
- Web Browsers and Web Crawlers
- Virus, Worm, Spyware, Trojan Horse and Phishing

- Antivirus Policies
 - Encryption and Decryption
 - Public and Private Keys
 - RSA
 - Digital Signature
 - Password Security Policies
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Module II : Threat Detection and Device Security Management

- Intrusion Detection System (IDS)
 - Intrusion Prevention System (IPS)
 - Social Media Marketing
 - Social Media Security
 - Importance of Apple Security
 - Basics of Networking
 - Windows Security
 - Smartphone Block Diagram
 - Desktop Security
 - Mobile Security
 - Wi-Fi Security
 - Stateful and Stateless Firewall
 - Linux and Windows Firewall
 - Cyber Security Acts and Laws in India
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Module III : Strategic Management of E-Commerce and Digital Payment

- HTTP and HTTPS
- E-Commerce and M-Commerce
- Internet and Mobile Banking Security
- Rules for Debit and Credit Card Usage
- UPI Security Guidelines
- Mobile Wallet and E-Wallet
- Digital Forensics
- Governance, Risk and Compliance
- RBI Guidelines for Digital Payments
- Customer Protection